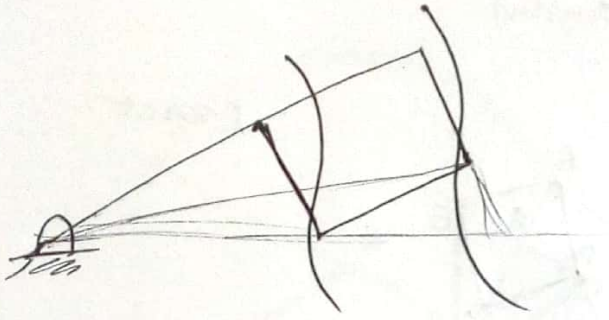
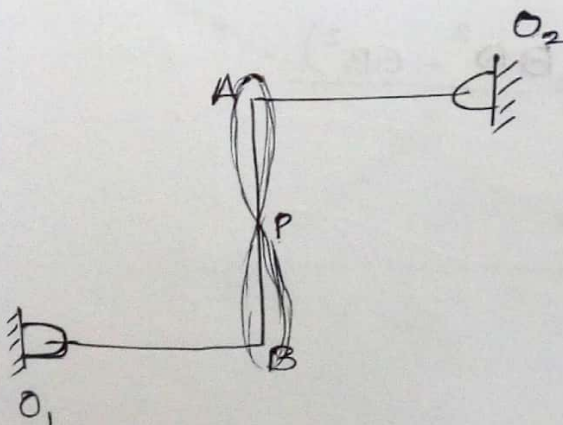
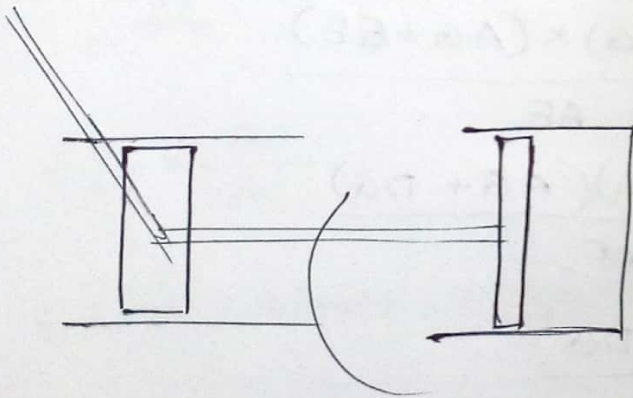


Common Mechanisms.



Straight Line Mechanisms
to guide piston in steam engines.

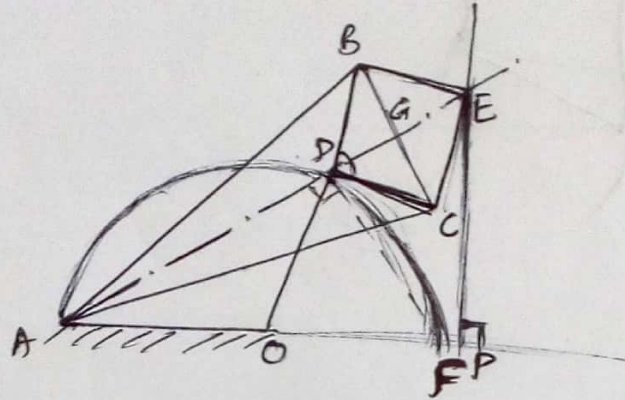
Watt Mechanism.



Approximate straight
line mechanism

Pomcellius Mechanism

Exact



$$AO = OD$$

$$AB = AC$$

$$DB = BE = EC = DC$$

$\triangle ADF$ and $\triangle APE$

$$\frac{AP}{AE} = \frac{AD}{AF}$$

$$AP = \frac{AD \times AE}{AF}$$

$$= \frac{(AG - DG) \times (AG + GE)}{AF}$$

$$= \frac{(AG - DG)(AG + DG)}{AF}$$

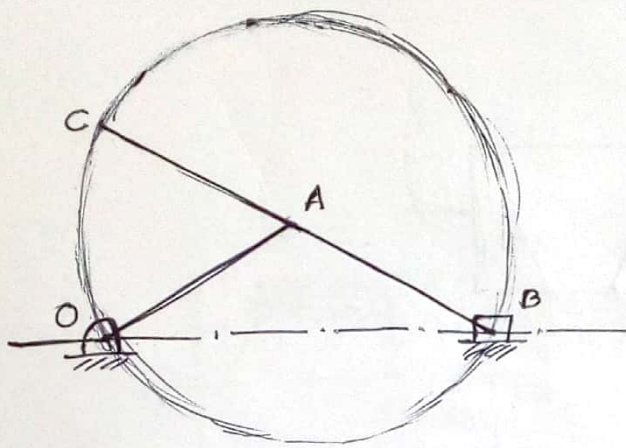
$$= \frac{AG^2 - DG^2}{AF}$$

$$= \frac{AB^2 - GB^2 - (BD^2 - GB^2)}{AF}$$

$$= \frac{AB^2 - BD^2}{AF}$$

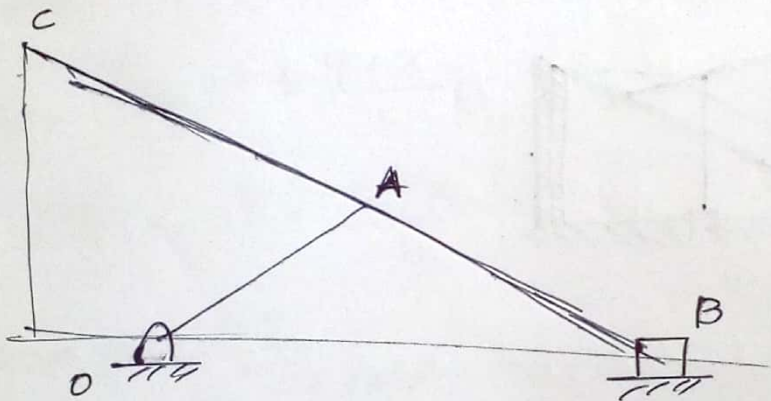
Scott Russel Mechanism

Exact straight line mechanism



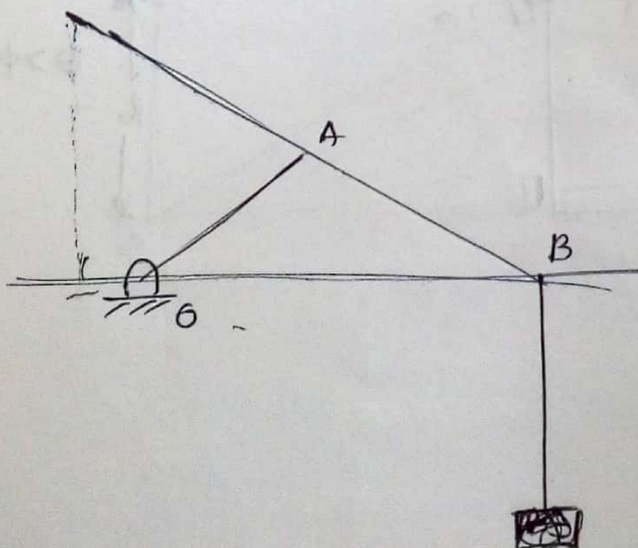
Modified Scott Russel Mechanism

Approximate straight line mechanism

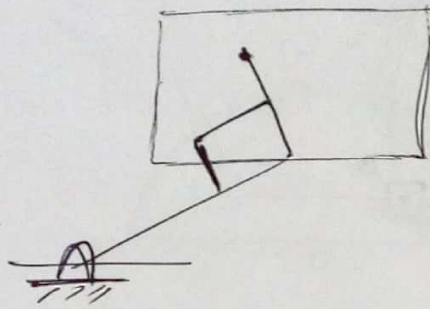


$$\frac{OA}{AB} = \frac{AB}{AC}$$

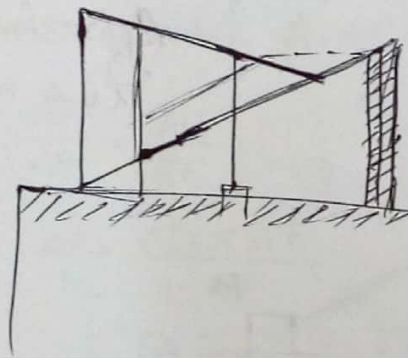
Grasshopper Mechanism



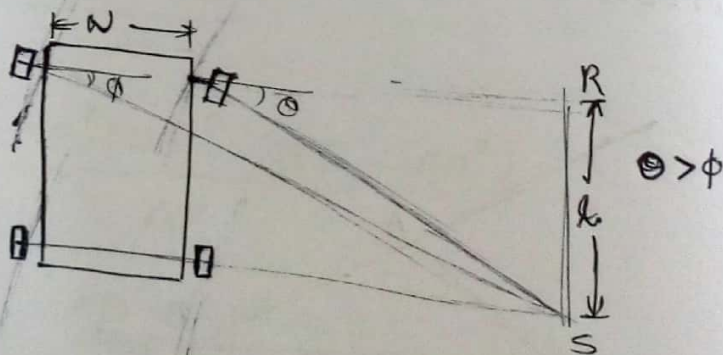
Engine Indicator



Simpler Indicator



Steering Mechanism

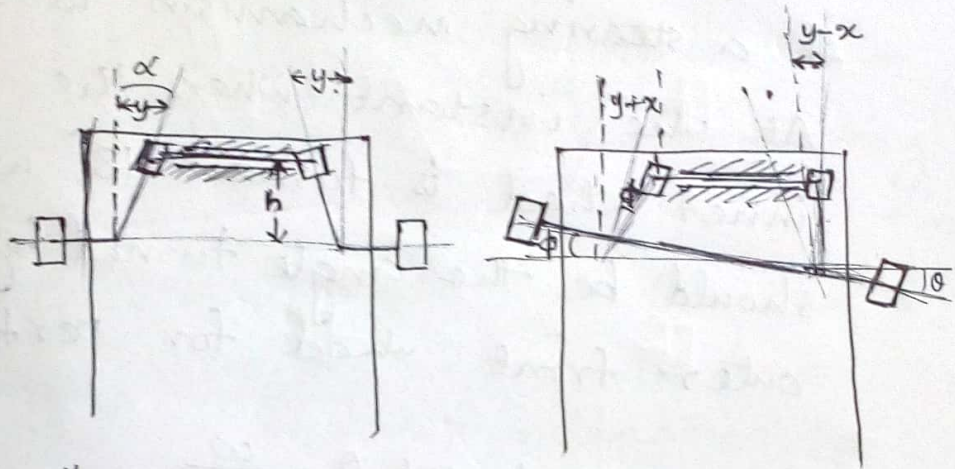


$$\cot \theta = \frac{QR}{RS}$$

$$\cot \phi = \frac{PR}{RS}$$

$$\cot \phi - \cot \theta = \frac{PR - QR}{RS} = \frac{PQ}{RS} = \frac{w}{R}$$

Davis' steering mechanism



$$\tan \alpha = \frac{y}{h}$$

$$\tan (\alpha + \phi) = \frac{y+x}{h}$$

$$\tan (\alpha - \theta) = \frac{y-x}{h}$$

$$\tan \theta = \frac{hx}{y^2 - xy + b^2}$$

$$\tan \phi = \frac{hx}{y^2 + xy + b^2}$$

$$\cot \phi - \cot \theta = \frac{y^2 + xy + b^2}{hx} - \left(\frac{y^2 - xy + b^2}{hx} \right)$$

$$= \frac{2xy}{hx}$$

$$= \frac{2y}{h}$$

$$= \frac{w}{l}$$

The ratio between the widths of the front axle and that of the wheel base of a steering mechanism is 0.44.

At the instant when the front inner wheel is turned 18° what should be the angle turned by the outer front wheel for perfect steering

$$\cot \phi - \cot \theta = \frac{w}{l}$$

$$\cot \phi - \cot 18 = 0.44$$

$$\phi = 15.86$$

The distance between the steering pivots of a Daves steering is 1.3 m. The wheel base is 2.75 m. What will be the inclination of crank arms to the longitudinal axis of the vehicle when it is moving in a straight path.

$$\tan \alpha = \frac{y}{b}$$

$$= \frac{0.65}{2.75}$$

$$\alpha = 13.298^\circ$$

$$w = 2y$$

$$y = 1.3/2$$

$$= 0.65 \text{ m}$$

The crank arm of a Daves steering is at a distance of 192 mm from the front main axle where as the difference between their lengths is 96 mm. If the distance between the steering pivots of the main axle is 1.4 m. Determine the length of the chassis between the front and rear base wheels (l). Also find the inclination of crank arms to the longitudinal axis of vehicle.

$$h = 192 \text{ mm}$$

$$y = 96/2$$

$$= 48 \text{ mm}$$

$$w = 1.4 \text{ m}$$

$$\frac{2y}{h} = \frac{w}{l}$$

$$l = \frac{wh}{2y}$$

$$= \frac{1.4 \times 192}{0.096}$$

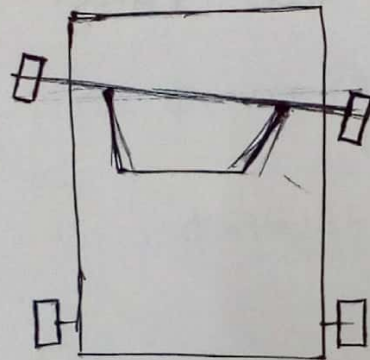
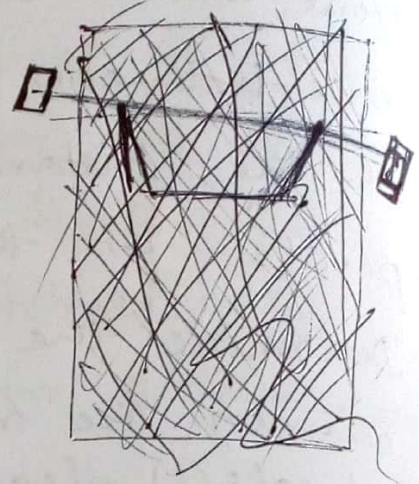
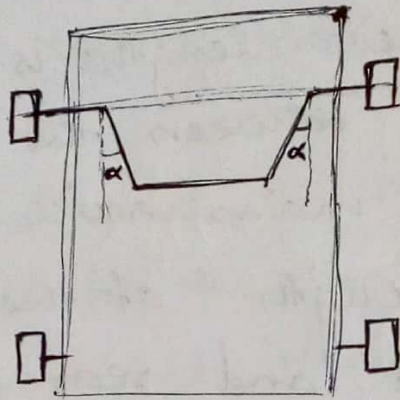
$$= 2.8 \text{ m}$$

$$\tan \alpha = \frac{y}{h}$$

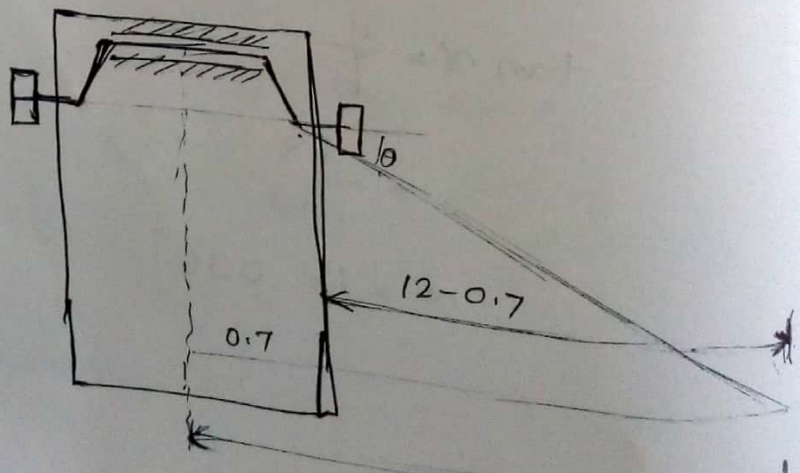
$$= \frac{48}{192}$$

$$\alpha = 14.036^\circ$$

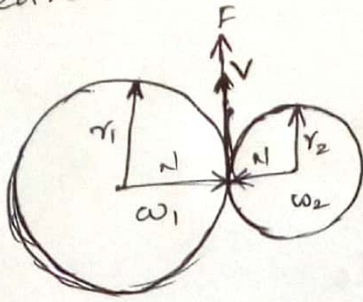
Ackermann Steering Mechanism



$\cot \phi - \cot \theta$ not a constant



Gears



$$V = r_1 \omega_1 = r_2 \omega_2$$

Pitch circle

Tooth thickness

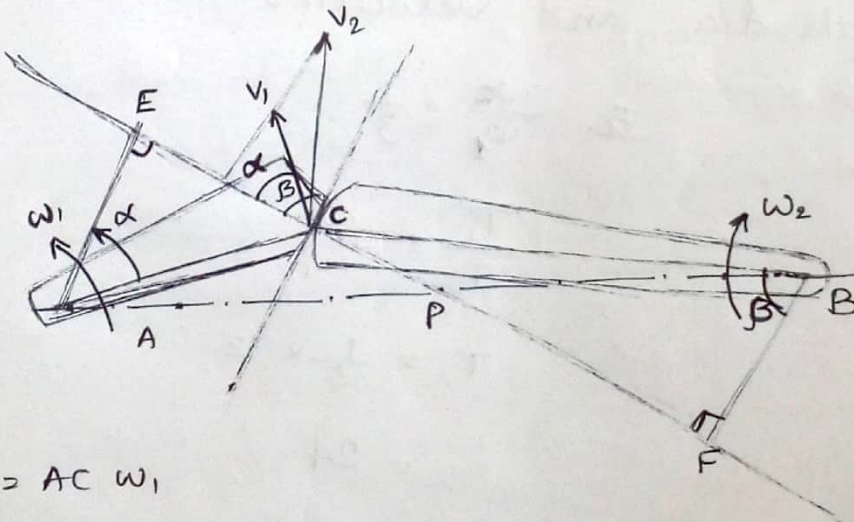
Tooth space

Tooth thickness = Tooth space

Addendum

Dedendum

Law of Gearing



$$V_1 = AC \omega_1$$

$$V_2 = BC \omega_2$$

$$V_1 \cos \alpha = AC$$

$$\omega_1 \cos \alpha = AE \omega_1$$

$$V_2 \cos \beta = BC$$

$$\omega_2 \cos \beta = BF \omega_2$$

$$AE \omega_1 = BF \omega_2$$

$$\frac{\omega_1}{\omega_2} = \frac{BF}{AE} = \frac{BP}{AP}$$

Conjugate Gears

$$\text{Circular Pitch} = \frac{\pi d}{T} \text{ mm}$$

T - Number of teeth

d - Dia of pitch circle

$$\text{Diametral Pitch} = \frac{T}{d} \text{ mm}^{-1}$$

$$\text{Module Pitch} = \frac{d}{T} \text{ mm}$$

$$\frac{d_1}{d_2} = \frac{T_1}{T_2} = \frac{\omega_2}{\omega_1} = \frac{N_2}{N_1}$$

Two spur gears having velocity ratio of $\frac{1}{3}$, the driven gear has 72 teeth of 8 mm module and rotates at 300 rpm. Calculate the number of teeth and the speed of the driver. What will be the pitch dia and velocities.

$$\frac{\omega_2}{\omega_1} = \frac{1}{3}$$

$$\frac{T_1}{T_2} = \frac{\omega_2}{\omega_1}$$

$$T_1 = \frac{1}{3} \times 72$$

$$= 24$$

$$N_1 = \frac{\omega_2}{\omega_1} \times N_2$$

$$= 3 \times 300$$

$$= 900 \text{ rpm}$$

$$d_1 = 8 \times 24$$
$$= 192 \text{ mm}$$

$$T_1 = 24$$

$$m = \frac{d}{T}$$

$$d = m T$$

$$r_1 = \frac{m T}{2}$$

$$= \frac{8 \times 24}{2}$$

$$= 96 \text{ mm}$$

$$\omega_1 = \frac{\pi D N_1}{60}$$

$$= \frac{2 \times 3.14 \times 900}{60}$$

$$= 94.2$$

$$V_1 = \omega_1 r_1$$

$$= 96 \times 94.2$$

$$=$$

$$T_2 = 72$$

$$r_2 = \frac{8 \times 72}{2}$$

$$= 288 \text{ mm}$$

$$\omega_2 = \frac{2\pi N_2}{60}$$

$$= \frac{2 \times 3.14 \times 300}{60}$$

$$= 31.4$$

$$V_2 = \omega_2 r_2$$

$$= 288 \times 31.4$$

$$=$$

The number of teeth of a spur gear is 30 and it rotates at 200 rpm. What will be the circular pitch and pitch dia velocity if it has a module of 2 mm.

$$T = 30$$

$$N = 200 \text{ rpm}$$

$$m = 2 \text{ mm}$$

$$d = m T$$

$$= 2 \times 30$$

$$= 60 \text{ mm}$$

$$\text{Circular pitch} = \frac{\pi d}{T}$$

$$= \frac{3.14 \times 60}{30}$$

$$= 6.28 \text{ mm}$$

$$\omega = \frac{2\pi N}{60}$$

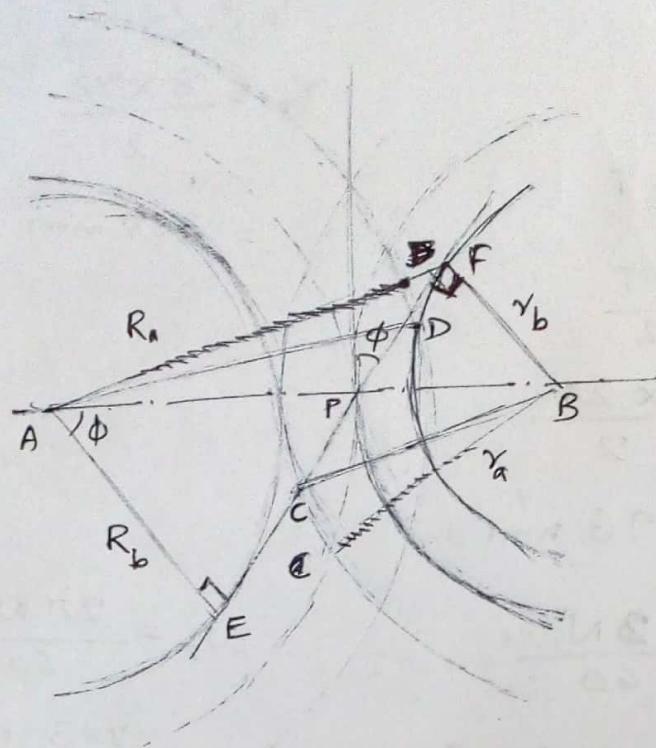
$$= \frac{2 \times 3.14 \times 200}{60}$$

$$= 21$$

$$V = \omega r$$

$$= 21 \times 30$$

$$= 62$$



$$R_b = R \cos \phi$$

$$r_b = r \cos \phi$$

$$PC = FC - FP = \sqrt{r_a^2 - r^2 \cos^2 \phi} - r \sin \phi$$

Path of approach

$$PD = \sqrt{R_a^2 - R^2 \cos^2 \phi} - R \sin \phi$$

Path of recess

Total path of

$$\text{Max. path of contact} = (R + r) \sin \phi$$

contact

Two involute gears in mesh have a

module of 8 mm and a pressure angle

of 20° . The larger gear has 57 teeth

while pinion has 23 teeth. If the

addenda on the pinion and gear

wheels are equal to 1 module find

the length of path of approach, length

of path of

and the path of

contact. The smaller gear is the driver

$$m = 8$$

$$D = mT$$

$$= 8 \times 57$$

$$= 456 \text{ mm}$$

$$d = mt$$

$$= 8 \times 23$$

$$= 184 \text{ mm}$$

$$R = 228 \text{ mm}$$

$$r = 92 \text{ mm}$$

$$R_a = R + a$$

$$= 228 + 8$$

$$= 236$$

$$r_a = r + a$$

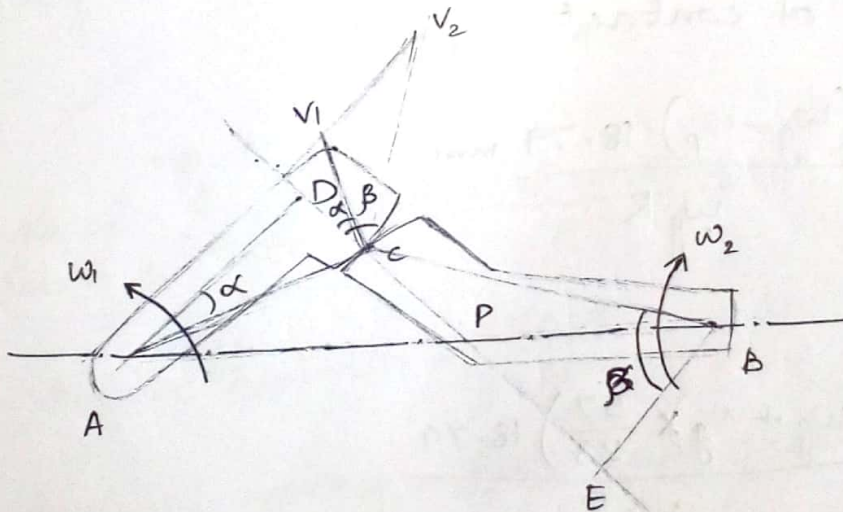
$$= 92 + 8$$

$$= 100$$

$$P.O.A. = \sqrt{(228)^2 - (92)^2} \cos 20^\circ - 92 \sin 20^\circ = 124.5 \text{ mm}$$

$$P.O.R = 179.5 \text{ mm}$$

$$\text{Path of contact} = 109.446 \text{ mm}$$



$$\frac{\omega_1}{\omega_2} = \frac{PB}{AP} = \frac{PE}{BP}$$

$$V_2 \sin \beta - V_1 \sin \alpha$$

$$= BC \omega_2 \frac{CE}{BC} - AC \omega_1 \frac{DC}{AC}$$

$$= \omega_2 CE - \omega_1 DC$$

$$= \omega_2 (PE + CP) - \omega_1 (PD - PC)$$

$$= \omega_2 CP + \omega_1 CP + \omega_2 PE - \omega_1 PD$$

$$\frac{\omega_1}{\omega_2} = \frac{PB}{AP} = \frac{PE}{BP}$$

Find the ratio of sliding to rolling velocity at the beginning of contact, at the pitch point, at the end of contact

Beginning

$$\text{Ratio} = \frac{(w_g + w_p) 20.9}{w_g R}$$

$$= \frac{(w_g + w_g \times \frac{57}{23}) 20.9}{w_g R}$$

$$\text{Ratio} = 0.31$$

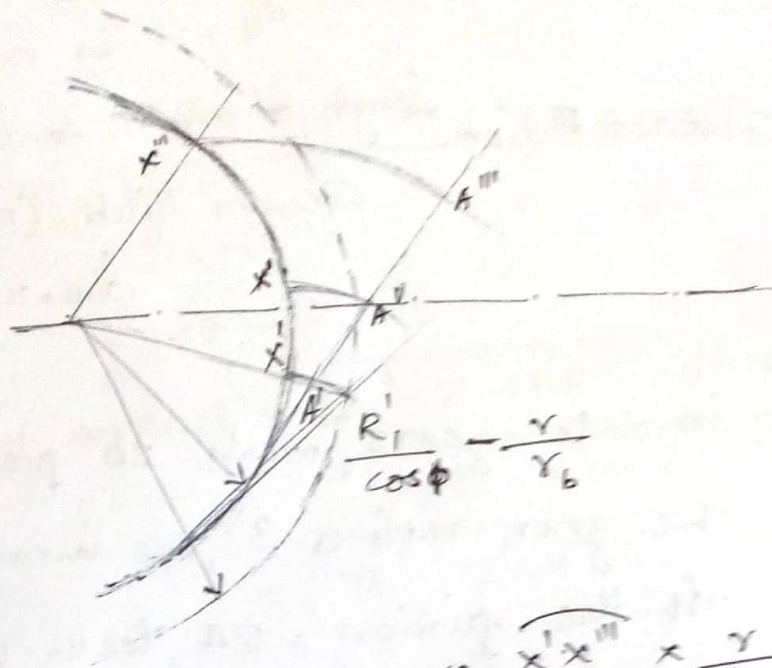
$$\text{Pitch} = 0$$

End of contact

$$\text{Ratio} = \frac{(w_g + w_p) 18.79 \text{ mm}}{w_g R}$$

$$= \frac{(w_g + w_g \times \frac{57}{23}) 18.79}{w_g R}$$

$$= 0.27$$



$$\begin{aligned} \text{Arc of contact} &= \text{Arc } X'X''' \times \frac{r}{\gamma_b} \\ &= \frac{\text{Length of path of contact}}{\cos \phi} \end{aligned}$$

$$\begin{aligned} \text{Arc of contact} &= \frac{39.79}{\cos 20^\circ} \\ &= 42.343 \end{aligned}$$

$$\begin{aligned} r\theta &= l \\ \text{Angle of action, } \theta &= l/r = \frac{\text{Arc of contact}}{R} = \end{aligned}$$

$$\text{Arc of approach} = \frac{20.9}{\cos 20}$$

$$\text{Angle of approach} = \frac{\text{Arc of approach}}{r}$$

Contact Ratio : Average number of teeth pairs in contact

$$\text{Contact Ratio} = \frac{\text{Length of arc of contact}}{\text{Circular Pitch } (P_c)}$$

$\pi m = \pi d / T$

Two involute gears in mesh 20° pressure angle, the gear ratio is 3, the number of teeth on the pinion = 24, teeth have a module of 6 mm. The pitch circle velocity is 2.6 m/s. An the addendum = 1 module. Determine the angle of action of the pinion, angle turned by pinion when pair of teeth is in mesh and the maximum velocity of sliding.

$$t = 24$$

$$T = 24 \times 3$$

$$= 72$$

$$m = 6 \text{ mm}$$

$$r = \frac{mt}{2}$$

$$= \frac{6 \times 24}{2}$$

$$= 72$$

$$R = \frac{mT}{2}$$

$$= \frac{6 \times 72}{2}$$

$$= 216$$

$$\begin{aligned} r_a &= r + a \\ &= 72 + 6 \\ &= 78 \text{ mm} \end{aligned}$$

$$\begin{aligned} R_a &= R + a \\ &= 216 + 6 \\ &= 222 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Total Path of contact} &= \sqrt{r_a^2 - r^2 \cos^2 \phi} - r \sin \phi \\ &\quad + \sqrt{R_a^2 - R^2 \cos^2 \phi} - R \sin \phi \\ &= \sqrt{78^2 - 72^2 \cos^2 20} - 72 \sin 20 \\ &\quad + \sqrt{222^2 - 216^2 \cos^2 20} - 216 \sin 20 \\ &= \cancel{217.9} + \cancel{2.87} \quad 14.19 + 16.04 \\ &= \cancel{5.66} \quad 30.23 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Arc of contact} &= \frac{\text{Path of contact}}{\cos \phi} \\ &= \frac{30.23}{\cos 20} \\ &= 32.17 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Angle of action, } \theta &= l/r \\ &= \frac{32.17}{72} \\ &= 0.447 \text{ radian} \\ &= 26^\circ 48' \end{aligned}$$

$$\begin{aligned} \text{Contact Ratio} &= \frac{\text{Length of arc of contact}}{\pi m} \\ &= \frac{32.17}{3.14 \times 6} \\ &= 0.17 \end{aligned}$$

$$\begin{aligned} \omega_p &= \frac{1.5 \text{ m/s}}{r} \\ &= \frac{1.5}{72} = \end{aligned}$$

Two 20° gears have a module pitch of 4mm. The number of teeth on gears 1 and 2 are 40 and 24 respectively. The gear 2 rotates at 600 rpm. Find velocity of sliding when the contact is at the tip of the tooth of gear 2. Take addendum = 1 module. Also find the maximum velocity of sliding.

~~At the tip~~

$$\phi = 20^\circ$$

$$m = 4 \text{ mm}$$

$$t = 24$$

$$T = 40$$

$$r = \frac{m \cdot t}{2}$$

$$= \frac{4 \times 24}{2}$$

$$= 48 \text{ mm}$$

$$r_a = 48 +$$

$$r_a = r + a$$

$$= 48 + 4$$

$$= 52 \text{ mm}$$

$$R = \frac{m \cdot T}{2}$$

$$= \frac{4 \times 40}{2}$$

$$= 80 \text{ mm}$$

$$R_a = R + a$$

$$= 80 + 4$$

$$PC = \sqrt{r_a^2 - r^2 \cos^2 \phi} - r \sin \phi$$

$$= \sqrt{52^2 - 48^2 \cos^2 20^\circ} - 48 \sin 20^\circ$$

$$= 9.46 \text{ mm}$$

$$N = 600 \text{ rpm}$$

$$\omega_2 = \frac{2\pi N}{60}$$

$$= \frac{2 \times 3.14 \times 600}{60}$$

$$= 62.8$$

$$\omega_1 = \omega_2 \times \frac{T_2}{T_1}$$

$$= 62.8 \times \frac{24}{40}$$

$$= 37.68$$

At the tip

$$\text{Velocity} = (\omega_1 + \omega_2) PC$$

$$= (62.8 + 37.68) 9.46$$

$$= 950.54 \text{ mm/s}$$

$$PB = \sqrt{R_a^2 - R^2 \cos^2 \phi} - R \sin \phi$$

$$= \sqrt{84^2 - 80^2 \cos^2(20^\circ)} - 80 \sin(20^\circ)$$

$$= 10.12 \text{ mm}$$

$$\text{Maximum velocity} = (\omega_1 + \omega_2) PB$$

$$= (62.8 + 37.68) 10.12$$

$$= 1016.86 \text{ mm/s}$$

Two gears having diameter
Each of 2 gears in a mesh have 48
teeth and a module of 8 mm, $\phi = 20^\circ$,
a ~~teeth~~ involute profile.

The arc of contact is 2.25 times the
circular pitch. Determine the addendum.

$$t = 48$$

$$m = 8 \text{ mm}$$

$$\phi = 20^\circ$$

$$\frac{\text{Arc of contact}}{\text{Circular pitch}} = 2.25$$

$$\text{Circular pitch} = \pi m$$

$$= 3.14 \times 8$$

$$= 25.12 \text{ mm}$$

$$\text{Arc of contact} = 2.25 \times 25.12$$

$$= 56.52 \text{ mm}$$

$$\text{Path of contact} = \text{Arc of contact} \times \cos \phi$$

$$= 56.52 \times \cos 20$$

$$= 53.11 \text{ mm}$$

$$r = \frac{mt}{2}$$

$$= \frac{8 \times 48}{2}$$

$$= 192 \text{ mm}$$

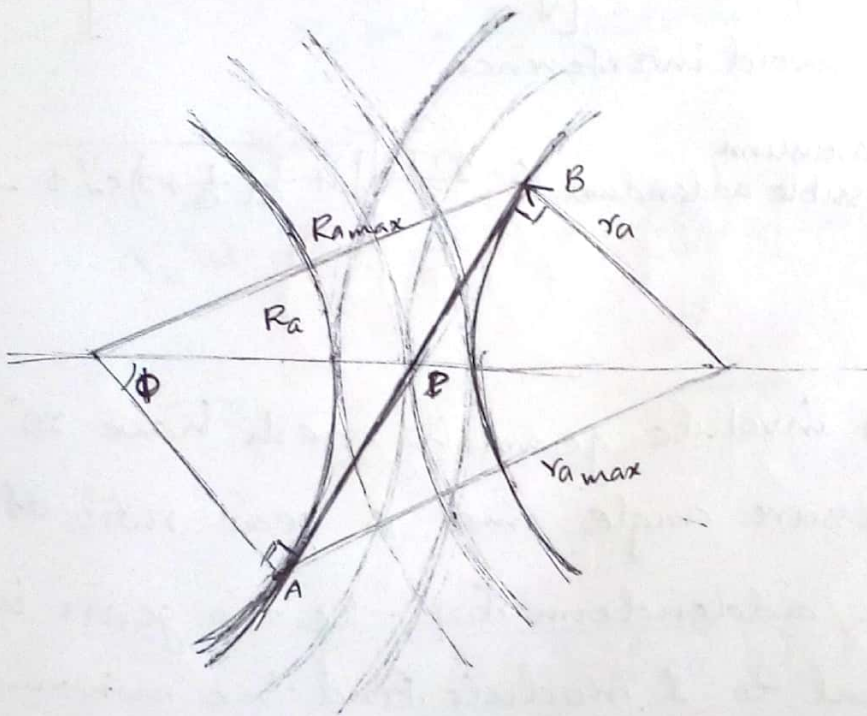
$$2 \times \left(\sqrt{r_a^2 - r^2 \cos^2 \phi} - r \sin \phi \right) = 53.11$$

$$2(\sqrt{r_a^2 - 192^2 \cos^2 20} - 192 \sin 20) = 53.11$$

$$\sqrt{r_a^2 - 192^2 \cos^2 20} = 92.22$$

$$r_a^2 = \frac{41056.789}{82643.45}$$

$$r_a = \cancel{180} \rightarrow 202 \text{ mm}$$



$$\frac{\gamma}{R} = \frac{t}{T} = \frac{1}{G}$$

$$R = \frac{R_b}{\cos \phi}$$

$$AP = R \sin \phi$$

$$PB = r \sin \phi$$

$$AP + PB = (R + r) \sin \phi$$

$$R_{\max} = \sqrt{R^2 \cos^2 \phi + (r+R)^2 \sin^2 \phi}$$

$$= \sqrt{R^2 \cos^2 \phi + (r^2 + R^2 + 2rR) \sin^2 \phi}$$

$$= \sqrt{R^2 (\cos^2 \phi + \sin^2 \phi) + (y^2 + 2yR) \sin^2 \phi}$$

$$= \sqrt{R^2 + (r^2 + 2rR) \sin^2 \phi}$$

$$= R \sqrt{1 + \left(\frac{r^2}{R^2} + \frac{2r}{R} \right) \sin^2 \phi}$$

$$= R \sqrt{1 + \frac{r}{R} \left(\frac{r}{R} + 2 \right) \sin^2 \phi}$$

$$= R \sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi}$$

$$a_{w \max} = R \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1 \right]$$

$$a_p \max = r \left[\sqrt{1 + G(G+2) \sin^2 \phi} - 1 \right]$$

To avoid interference

$$\text{Maximum possible addendum} \leq \frac{mT}{2} \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1 \right]$$

Two involute gears in mesh have 20° pressure angle and a gear ratio of 1. The addendum for the two gears is equal to 1 module. Find the minimum number of teeth required to avoid interference.

$$1m \leq \frac{mT}{2} \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1 \right]$$

$$T \geq \frac{2}{\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1}$$

$$\geq \frac{2}{\sqrt{1 + \frac{1}{1} \left(\frac{1}{1} + 2 \right) \sin^2 20^\circ} - 1}$$

$$\geq \frac{2}{\sqrt{1 + 3 \times 0.117} - 1} = 12.32$$

Two 20° involute spur gears mesh externally and gives a velocity ratio of 3.

The module is 3mm and both the gear and pinion have an addendum coefficient of 1.1. If the pinion rotates at 120 rpm

Determine (a) The minimum number of teeth of each wheel to avoid interference (b) the contact ratio. (c) Addendum for the ~~above~~ ^{wheel} gears ~~is~~ ^{is} 51. (d) If $\phi = 22.5^\circ$, what is α_w of module when

$$G = 3$$

$$\alpha_w m = a_w, \text{ Addendum}$$

α_w - Addendum coefficient

$$\alpha_w m \leq \frac{m T}{2} \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1 \right]$$

$$T \geq \frac{2 \alpha_w}{\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1}$$

$$\geq \frac{2 \times 1.1}{\sqrt{1 + \frac{1}{3} \left(\frac{1}{3} + 2 \right) \sin^2 20^\circ} - 1}$$

$$\geq \frac{2.2}{\sqrt{1 + \frac{1}{3} \times 0.117} - 1}$$

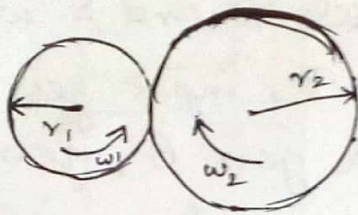
$$\geq 49.44$$

$$T = 51$$

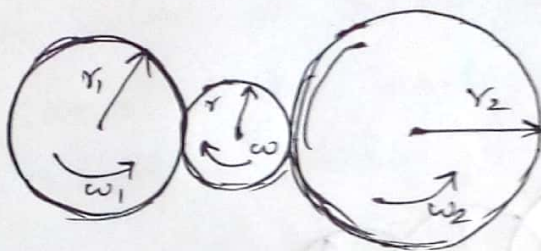
$$t = \frac{T}{G} = \frac{51}{3}$$

$$= 17$$

Gear Trains



$$\frac{\omega_1}{\omega_2} = \frac{r_2}{r_1}$$

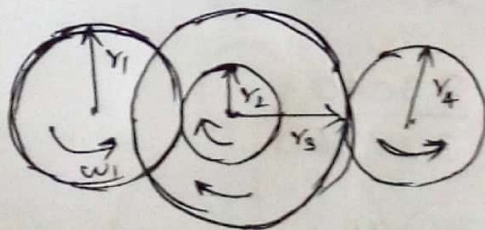


$$\frac{\omega_1}{\omega_3} = \frac{r_2}{r_1}$$

$$\frac{\omega}{\omega_2} = \frac{r_2}{r}$$

$$\frac{\omega_1}{\omega} \times \frac{\omega}{\omega_2} = \frac{r_2}{r_1} \times \frac{r}{r_2}$$

$$\frac{\omega_1}{\omega_2} = \frac{r_2}{r_1}$$

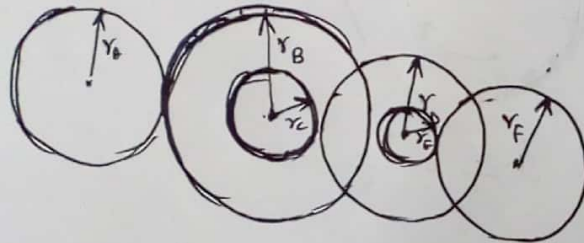


$$\frac{\omega_2}{\omega_1} = \frac{r_1}{r_2}$$

$$\frac{\omega_2}{\omega_1} \times \frac{\omega_4}{\omega_3} = \frac{r_3}{r_4} \times \frac{r_1}{r_2}$$

$$\frac{\omega_2}{\omega_1} = \frac{r_3}{r_4}$$

In a compound gear train, a compound set B and C with number of $T_B = 56$ and $T_C = 30$. Another compound set D and E with $T_D = 80$ and $T_E = 32$. The input gear A having $T_A = 24$ meshes with gear B, gear C meshes with gear D and gear E meshes with the output gear F having $T_F = 72$. Find the speed of gear F if A is rotating at 800 rpm.

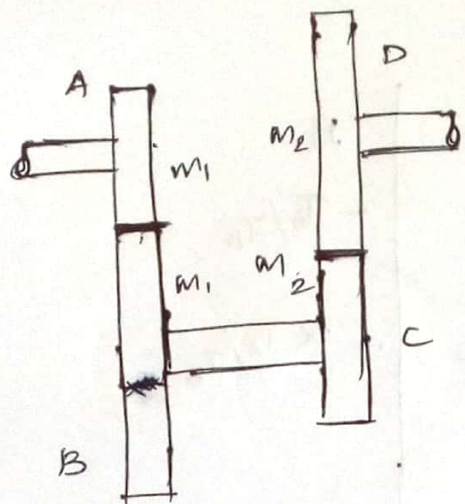


$$\begin{aligned}\frac{\omega_F}{\omega_A} &= \frac{N_F}{N_A} = \frac{T_A * T_C * T_E}{T_B * T_D * T_F} \\ &= \frac{T_A * T_C * T_E}{T_B * T_D * T_F} \\ &= \frac{24 \times 30 \times 32}{56 \times 80 \times 72} \\ &= \frac{23040}{322560} \\ &= 0.071\end{aligned}$$

$$N_F = \cancel{800} \times 800 \times 0.071$$

$$\cancel{800 \times 0.071}$$

$$= 56.8$$



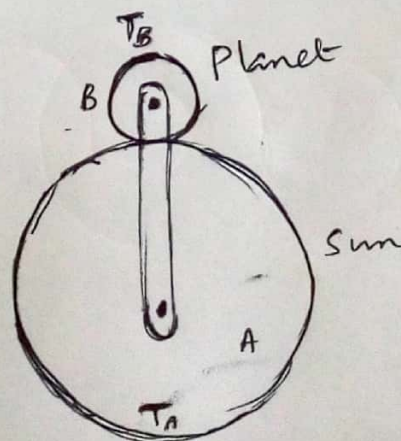
Reverted gear train

Design a reverted gear train to get a total train value of 12. The centre distance between the axes of the input gear and compound gears should be nearly 160 mm.

Train value is Ratio of speed of driven by speed of driving gear

$$\text{Speed ratio} = \frac{1}{\text{Train value}}$$

Epicyclic Gear Train



	Arm	A	B
Arm fixed	-	1	$-T_A/T_B$
Arm fixed + A x rotation	-	x	$-x T_A/T_B$
y rotation for entire rotation	y	$x+y$	$y - (x T_A/T_B)$

Two meshing spur gears with 20° pressure angle having module of 4 mm. The centre distance is 220 mm and the number of teeth on the pinion is 40. To what distance the central distance is increased so that the pressure angle is increased to